

Application to the Picard group

SGA 2, Exposé XI

English draft

A. Grothendieck

About this draft

This is an unofficial English translation of Exposé XI, “Application to the Picard group”, from SGA 2. The source volume is A. Grothendieck (notes by a group of auditors), with an exposé by M. Raynaud, *Cohomologie locale des faisceaux cohérents et théorèmes de Lefschetz locaux et globaux* (SGA 2), Séminaire de géométrie algébrique du Bois Marie, 1962. It follows the slightly corrected SMF recomposition, arXiv:math/0511279.

This draft contains the assigned body of the source, including proofs, original footnotes, and editor notes (N.D.E.). Statement numbering and mathematical notation follow the source. References to material outside this draft retain their original numbers. Apparent mathematical misprints in the source have been retained; they are recorded in the accompanying README, separately from the translated text. Scholarly proofreading remains outstanding.

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EXPOSÉ XI

Application to the Picard group

This exposé is modelled on the preceding one, but this time the result of no. 1 is weaker.

Throughout this exposé, X will denote a locally noetherian prescheme, $\mathcal{I}$ a quasi-coherent ideal of $\mathcal{O}_X$, ($Y = V(\mathcal{I})$ is therefore a closed subset of X), U a variable open neighbourhood of Y in X , and $\hat{X}$ the formal completion of X along Y . For every ringed space $(Z, \mathcal{O}_Z)$, one denotes by $\mathbf{P}(Z)$ the category of invertible $\mathcal{O}_Z$ -Modules, in other words locally free of rank 1, and by $\text{Pic}(Z)$ the group of isomorphism classes of invertible Modules on Z .

1. Comparison of $\text{Pic}(\hat{X})$ and of $\text{Pic}(Y)$

For every $n \in \mathbf{N}$, set $X_n = (Y, \mathcal{O}_X/\mathcal{I}^{n+1})$ and $P_n = \mathcal{I}^{n+1}/\mathcal{I}^{n+2}$. The sequence of sheaves of abelian groups on Y

$$(1) \quad 0 \longrightarrow P_n \xrightarrow{u} \mathcal{O}_{X_{n+1}}^* \xrightarrow{v} \mathcal{O}_{X_n}^* \longrightarrow 1$$

is *exact*. Let us specify that the group structure of P_n is the additive structure, that $u(x) = 1 + x$ for every $x \in P_n$, and that v is the homomorphism deduced from the injection $\mathcal{I}^{n+2} \rightarrow \mathcal{I}^{n+1}$. One sees that v is surjective by remarking that, for every $y \in Y$, $\mathcal{O}_{X_n, y}$ is a local ring, quotient of $\mathcal{O}_{X_{n+1}, y}$ by a nilpotent ideal; the rest is equally trivial. One deduces from (1) an exact cohomology sequence:

$$(*) \quad H^1(Y, P_n) \xrightarrow{u^1} H^1(Y, \mathcal{O}_{X_{n+1}}^*) \xrightarrow{v^1} H^1(Y, \mathcal{O}_{X_n}^*) \xrightarrow{d} H^2(Y, P_n).$$

Moreover, for every $n \in \mathbf{N}$, one knows how to identify $\text{Pic}(X_n)$ and $H^1(Y, \mathcal{O}_{X_n}^*)$; moreover, if E is an invertible $\mathcal{O}_{X_{n+1}}$ -Module, corresponding to a cohomology class $c(E)$, the cohomology class corresponding to the inverse image of E on X_n is equal to $v^1(c(E))$. Hence the following proposition:

PROPOSITION 1.1. *Let us keep the notations introduced above. Let $p \in \mathbf{N}$. The map $\text{Pic}(\hat{X}) \rightarrow \text{Pic}(Y_n)$:*

- (i) *is injective for $n \geq p$, if $H^1(Y, P_n) = 0$ for $n \geq p$;*
- (ii) *is an isomorphism for $n \geq p$, if $H^i(Y, P_n) = 0$ for $n \geq p$ and $i = 1, 2$.*

Of course, the exact sequence $(*)$ contains more information than the proposition above. The reader will have noticed that nothing has been said about the functor $\mathbf{P}(\hat{X}) \rightarrow \mathbf{P}(Y)$. Given two invertible $\mathcal{O}_{\hat{X}}$ -Modules E, F , $H = \underline{\text{Hom}}(E, F)$ is also invertible. If one indicates by a subscript n the reduction modulo $\mathcal{I}^{n+1}$, one finds an exact sequence:

$$0 \longrightarrow H_0 \otimes P_n \longrightarrow \underline{\text{Hom}}(E_{n+1}, F_{n+1}) \longrightarrow \underline{\text{Hom}}(E_n, F_n) \longrightarrow 0.$$

Hence an exact cohomology sequence which we shall not write and whose interpretation is evident; one may use this remark to study the functor $\mathbf{P}$.

2. Comparison of $\text{Pic}(X)$ and of $\text{Pic}(\hat{X})$

The reader will find in exposé ??, no. ??, the proof of what follows:

PROPOSITION 2.1. *Suppose that one has $\text{Lef}(X, Y)$; then for every open neighbourhood U of Y in X , the functor*

$$(2) \quad \mathbf{P}(U) \longrightarrow \mathbf{P}(\hat{X})$$

is fully faithful, the map

$$(3) \quad \mathrm{Pic}(U) \longrightarrow \mathrm{Pic}(\widehat{X})$$

is therefore injective. If one has $\mathrm{Leff}(X, Y)$, the map (4) is an isomorphism:

$$(4) \quad \varinjlim_U \mathrm{Pic}(U) \longrightarrow \mathrm{Pic}(\widehat{X}).$$

COROLLARY 2.2. *Suppose that one has $\mathrm{Lef}(X, Y)$ and that for every integer $n \geq p$, one has $H^1(Y, P_n) = 0$; then for every open $U \supset Y$, the maps*

$$\mathrm{Pic}(X) \longrightarrow \mathrm{Pic}(U) \longrightarrow \mathrm{Pic}(Y_n)$$

are injective if $n \geq p$. If one has $\mathrm{Leff}(X, Y)$ and if moreover, for every integer $n \geq p$, one has $H^i(Y, P_n) = 0$ for $i = 1$ and $i = 2$, then the map

$$\varinjlim_U \mathrm{Pic}(U) \longrightarrow \mathrm{Pic}(Y_n)$$

is an isomorphism for $n \geq p$.

3. Comparison of $\mathbf{P}(X)$ and of $\mathbf{P}(U)$

A definition:

DEFINITION 3.1¹. *Let X be a prescheme and let Z be a closed subset of X . Set $U = X - Z$. One says that X is *parafactorial* at the points of Z if, for every open V of X , the functor $\mathbf{P}(V) \rightarrow \mathbf{P}(V \cap U)$ is an equivalence of categories. One also says that the pair (X, Z) is *parafactorial*.*

Recall that $\mathbf{P}(Z)$ denotes the category of Modules locally free of rank 1 on Z .

DEFINITION 3.2. A noetherian local ring is said to be *parafactorial* if the pair $(\mathrm{Spec}(A), \{\mathfrak{r}(A)\})$ is parafactorial.

One proves the following proposition which shows that the notion is “pointwise”:

PROPOSITION 3.3. *Suppose X locally noetherian. In order that the pair (X, Z) be parafactorial, it is necessary and sufficient that, for every $z \in Z$, the local ring $\mathcal{O}_{X,z}$ be so.*

Let us remark that in parafactorial there is “fully faithful”. One proves as in lemma ?? of exposé ?? the:

LEMMA 3.4. *If X is a locally noetherian prescheme and if $Z = X - U$ is a closed subset of X , the following conditions are equivalent:*

- (i) *for every open V of X , the functor $\mathbf{P}(V) \rightarrow \mathbf{P}(V \cap U)$ is fully faithful;*
- (ii) *the homomorphism $\mathcal{O}_X \rightarrow i_*(\mathcal{O}_U)$ is an isomorphism;*
- (iii) *for every $z \in Z$, one has $\mathrm{prof}(\mathcal{O}_{X,z}) \geq 2$.*

Thus “parafactorial” means that the conditions of 3.4 are satisfied and that, for every open V of X , the homomorphism $\mathrm{Pic}(V) \rightarrow \mathrm{Pic}(V \cap U)$ is surjective. In particular, if X is the spectrum of a noetherian local ring, one finds:

PROPOSITION 3.5. *Let A be a noetherian local ring; in order that it be parafactorial it is necessary and sufficient that $\mathrm{prof} A \geq 2$ and $\mathrm{Pic}(X' - \{x\}) = 0$, where one has set $X' = \mathrm{Spec}(A)$ and where x is the unique closed point of X' .*

One will note that a local ring of dimension ≤ 1 is never parafactorial since its depth is ≤ 1 . Thus *factorial does not imply parafactorial*; it is however true for noetherian local rings of dimension ≥ 2 , as we shall see below.

¹For a more detailed study of the notion of parafactoriality, and the proof of 3.3, cf. EGA IV 21.13, 21.14.

LEMMA 3.6. *Let X be a locally noetherian prescheme and let Z be a closed subset of X . Let $f: X_1 \rightarrow X$ be a faithfully flat and quasi-compact morphism. Set $Z_1 = f^{-1}(Z)$. If (X_1, Z_1) is parafactorial, then (X, Z) is so.*

One remarks first that, if $i: (X - Z) \rightarrow X$ denotes the canonical immersion of $U = X - Z$ into X , the formation of the direct image by i of a quasi-coherent $\mathcal{O}_U$ -Module commutes with the base change f , since the latter is flat. It is therefore equivalent to assume the equivalent conditions of lemma 3.5 for (X, Z) or for (X_1, Z_1) , since f is a descent morphism for the category of quasi-coherent sheaves. It remains to prove that, for every open V of X , $\text{Pic}(V) \rightarrow \text{Pic}(V \cap U)$ is surjective. One makes the base change $V \rightarrow X$, which changes nothing (*sic*), and one is reduced to $V = X$. One then remarks that, if L is an invertible $\mathcal{O}_U$ -Module and if L admits a locally free embedding, this extension is isomorphic to $i_*(L)$, by what has just been seen. It remains to prove that $i_*(L)$ is invertible. Using once more the fact that the direct image by i commutes with flat base change, and that “locally free of rank 1” is a property which descends by a faithfully flat and quasi-compact morphism, we are done.

COROLLARY 3.7. *Let A be a noetherian local ring; if $\hat{A}$ is parafactorial A is so.*

Do not believe that, if A is parafactorial, $\hat{A}$ is so².

Before tackling the statement of the main theorem of this no. , let us make the connection with the theory of divisors and the notion of factorial ring³.

Let X be a *noetherian* and *normal* prescheme. Let $Z^1(X)$ be the free abelian group generated by the $x \in X$ such that $\dim \mathcal{O}_{X,x} = 1$. The local ring of such a point is a discrete valuation ring. One will denote by v_x the corresponding normalized valuation. Let $K(X)$ be the ring of rational functions of X and let

$$p: K(X)^* \longrightarrow Z^1(X)$$

be the map which to every $f \in K(X)^*$ associates the 1-codimensional cycle:

$$(f) = \sum_{x \in X, \dim \mathcal{O}_{X,x} = 1} v_x(f) \cdot x.$$

The image of p is denoted $P(X)$ and one calls its elements *principal divisors*⁴. One sets

$$\text{Cl}(X) = Z^1(X)/P(X).$$

Let $Z'^1(X)$ be the subgroup of $Z^1(X)$ whose elements are the *locally principal divisors*. One knows that

$$\text{Pic}(X) \simeq Z'^1(X)/P(X),$$

and consequently $\text{Pic}(X)$ identifies with a subgroup of $\text{Cl}(X)$.

Let us remark that if U is a dense open of X , $K(X) \rightarrow K(U)$ is an isomorphism, and that if $\text{codim}(X - U, X) \geq 2$, i.e. if every $x \in X$ such that $\dim \mathcal{O}_{X,x} \leq 1$ belongs to U , the homomorphism $Z^1(X) \rightarrow Z^1(U)$ and consequently $\text{Cl}(X) \rightarrow \text{Cl}(U)$ is also an isomorphism. Finally, if every $x \in U$ is factorial, i.e. $\mathcal{O}_{X,x}$ is so, then $Z^1(U) = Z'^1(U)$ hence $\text{Pic}(U) \simeq \text{Cl}(U)$.

PROPOSITION 3.7.1. *Let X be a noetherian and normal prescheme. Let $(U_i)_{i \in I}$ be a family of opens of X such that:*

- a) *the U_i form a filter base*⁵;
- b) *if one sets $Y_i = X - U_i$, one has $\text{codim}(Y_i, X) \geq 2$ for every i ,*

²**N.D.E.** for a precise study of the relation between the factoriality of A and of its completion, see (Heitmann R., “Characterization of completions of unique factorization domains”, *Trans. Amer. Math. Soc.* **337** (1993), no. 1, p. 379–387).

³For the generalities which follow, cf. also EGA IV 21.

⁴In accordance with the terminology of EGA IV 21, we now prefer to reserve the name of “divisors” for “locally principal divisors” or “Cartier divisors”.

⁵**N.D.E.** i.e. a decreasing filtered family.

c) if $x \in U_i$ for every $i \in I$, then $\mathcal{O}_{X,x}$ is factorial.

Then one has an isomorphism:

$$\varinjlim_{i \in I} \text{Pic}(U_i) \xrightarrow{\sim} \text{Cl}(X).$$

Let us remark that b) implies that every $x \in X$ such that $\dim \mathcal{O}_{X,x} \leq 1$ belongs to U_i for every i . Hence the U_i are dense and moreover the homomorphism $Z^1(U_i) \rightarrow Z^1(X)$ is an isomorphism, as is $K(U_i) \rightarrow K(X)$. Hence $\text{Pic}(U) \subset \text{Cl}(U_i) \simeq \text{Cl}(X)$. To prove what one wishes, it therefore suffices to show that every $D \in Z^1(X)$ belongs to $Z^1(U_i)$ for a suitable i . It suffices to do this for the irreducible and positive “divisors”. Let therefore $x \in X$ such that $\dim \mathcal{O}_{X,x} = 1$. It suffices to prove that there exists an $i \in I$ such that $\overline{\{x\}}$ is locally principal at the points of U_i . Let $\mathcal{I}$ be the largest ideal of definition of the closed set $\overline{\{x\}}$. The set of points in a neighbourhood of which $\mathcal{I}$ is free is an open U . Now $U \supset \bigcap_{i \in I} U_i$ by c). If one sets $Y = X - U$, one has $Y \subset \bigcup_{i \in I} Y_i$ with $Y_i = X - U_i$, now Y is closed, hence admits a finite number of generic points, hence is contained in the union of a finite number of Y_i , hence in Y_j for some $j \in I$, since the U_i form a filter base. Hence $U \supset U_j$. $\square$

COROLLARY 3.8. *Let X be a noetherian and normal prescheme and let Y be a closed subset of codimension ≥ 2 . Suppose that, for every $p \in X - Y$, $\mathcal{O}_{X,p}$ is factorial, then*

$$\text{Pic}(X - Y) \longrightarrow \text{Cl}(X - Y) \longrightarrow \text{Cl}(X)$$

are isomorphisms.

COROLLARY 3.9. *Let A be a noetherian and normal local ring. Set $X' = \text{Spec}(A)$ and $x = \mathfrak{r}(A)$. In order that A be factorial it is necessary and sufficient that $\text{Pic}(X' - \{x\}) = 0$ and that $\mathfrak{p} \in X' - \{x\}$ imply that $A_{\mathfrak{p}}$ is factorial.*

Indeed, in order that A be factorial it is necessary and sufficient that $\text{Cl}(X') = 0$ ⁶.

COROLLARY 3.10. *Let A be a noetherian local ring of dimension ≥ 2 . Set $X' = \text{Spec}(A)$ and let $x = \mathfrak{r}(A)$. Set $X = X' - \{x\}$. The following conditions are equivalent:*

- (i) A is factorial;
- (ii) a) for every $y \in X$, $\mathcal{O}_{X,p}$ is factorial, and
b) A is parafactorial, i.e. $\text{prof } A \geq 2$ and $\text{Pic}(X) = 0$.

Before proving this corollary, let us state the

SERRE’S CRITERION FOR NORMALITY 3.11⁷. *Let A be a noetherian local ring. In order that A be normal, it is necessary and sufficient that*

- (i) for every prime ideal $\mathfrak{p}$ of A such that $\dim A_{\mathfrak{p}} \leq 1$, $A_{\mathfrak{p}}$ be normal,
- (ii) for every prime ideal $\mathfrak{p}$ of A such that $\dim A_{\mathfrak{p}} \geq 2$, one have $\text{prof } A_{\mathfrak{p}} \geq 2$.

Let us prove 3.10.

(i) $\Rightarrow$ (ii). Knowing that a localization of a factorial ring is again so, one has (ii) a). Moreover A is normal hence $\text{prof } A \geq 2$ since $\dim A \geq 2$ (3.11 (ii)). Finally A is parafactorial; indeed $\text{Pic}(X) \simeq \text{Cl}(X') = 0$ (cf. 3.9).

(ii) $\Rightarrow$ (i). Let us first prove that A is normal by applying Serre’s criterion. Since $\dim A \geq 2$, condition (i) of the criterion is among the hypotheses. Moreover, for every $\mathfrak{p} \in X$, $A_{\mathfrak{p}}$ is factorial, hence normal, hence of depth ≥ 2 , at least if $\dim A_{\mathfrak{p}} \geq 2$. Finally $\text{prof } A \geq 2$ by (ii) b). It remains to apply 3.9.

Let us summarize what precedes:

PROPOSITION 3.12. *Let X be a locally noetherian prescheme and let $\mathcal{I}$ be a quasi-coherent ideal of X . Let $Y = V(\mathcal{I})$. Let $p \in \mathbf{N}$. Suppose that:*

⁶**N.D.E.** see Bourbaki, *Commutative algebra* VII.1.4, cor. of th. 2 and VII.3.2, th. 1

⁷Cf. EGA IV 5.8.6.

- 1) One has $\text{Leff}(X, Y)$ (Exposé ??);
- 2) $H^i(X, \mathcal{I}^{n+1}/\mathcal{I}^{n+2}) = 0$ if $i = 1$ or 2 and if $n \geq p$;
- 3) for every open neighbourhood U of Y in X and every $x \in X - U$, the ring $\mathcal{O}_{X,x}$ is parafactorial.

Then, for every $n \geq p$, and every open neighbourhood U of Y , the homomorphisms

$$\begin{array}{ccc} \text{Pic}(X) & \xrightarrow{\quad} & \text{Pic}(X_n) \\ & \searrow \quad \swarrow & \\ & \text{Pic}(U) & \end{array}$$

are isomorphisms.

One knows some parafactorial rings:

- THEOREM 3.13. (i) (Auslander-Buchsbaum)⁸ A regular noetherian local ring is factorial (hence parafactorial if its dimension is ≥ 2).
- (ii) A noetherian local ring of dimension ≥ 4 which is a complete intersection is parafactorial.

COROLLARY 3.14 (Samuel's conjecture⁹). A noetherian local ring A which is a complete intersection and which is factorial in codimension ≥ 3 (i.e. $\dim A_{\mathfrak{p}} \leq 3$ implies that $A_{\mathfrak{p}}$ is factorial) is factorial.

Let us prove the corollary

One proceeds by induction on the dimension of A .

If $\dim A \leq 3$, A is factorial by hypothesis.

If $\dim A > 3$, by the induction hypothesis, remarking that a localization of a complete intersection is again so, all the localizations of A other than A are factorial. By theorem 3.13 (ii), A is parafactorial, hence factorial by 3.10.

Let us prove 3.13 (i) (following Kaplansky)¹⁰.

Let A be a regular noetherian local ring, set $\dim A = n$. If $n = 0$ or 1 , the result is known. Suppose $n \geq 2$, proceed by induction on n and suppose $n \geq 2$ and the theorem proved for rings of dimension $< n$. Set $X' = \text{Spec}(A)$ and $X = X' - \{x\}$, where $x = \mathfrak{r}(A)$. The localizations of A other than A are regular and of dimension $< n$, hence factorial. Moreover $\text{prof } A = \dim A \geq 2$. It therefore suffices to prove that $\text{Pic}(X) = 0$ (cor. 3.10). Let therefore L be an invertible $\mathcal{O}_X$ -Module, one knows that one can extend it to a coherent $\mathcal{O}_{X'}$ -Module L' . There exists a resolution of L' by free $\mathcal{O}_X$ -Modules:

$$0 \longleftarrow L' \longleftarrow L'_1 \longleftarrow \cdots \longleftarrow L'_n \longleftarrow 0,$$

since the cohomological dimension of A is finite. By restriction to X' one obtains a finite free resolution. It therefore suffices to prove the following lemma:

LEMMA 3.15. Let $(X, \mathcal{O}_X)$ be a ringed space and let L be a locally free $\mathcal{O}_X$ -Module which admits a finite resolution by modules free of finite type. Then $\det(L) \simeq \mathcal{O}_X$.

⁸N.D.E. to be compared with the following purity result, due to Gabber. Let X be the spectrum of a regular local ring A of dimension 3, a of non-zero differential, i.e. $a \in \mathfrak{m} - \mathfrak{m}^2$ and U the complement of $V(a)$. Then a vector bundle on U is free (for a simple proof, see (Swan R.G., "A simple proof of Gabber's theorem on projective modules over a localized local ring" *Proc. Amer. Math. Soc.* **103** (1988), no. 4, p. 1025-1030). The rank 1 case is a particular case of theorem 3.13. For purity results concerning vector bundles of arbitrary rank, whether in the analytic or algebraic setting, see (Gabber O., "On purity theorems for vector bundles", *Internat. Math. Res. Notices* (2002), no. 15, p. 783-788).

⁹N.D.E. for a proof in the same vein, but more elementary, see Call F. & Lyubeznik G., "A simple proof of Grothendieck's theorem on the parafactoriality of local rings", in *Commutative algebra: syzygies, multiplicities, and birational algebra* (South Hadley, MA, 1992), Contemp. Math., vol. 159, American Mathematical Society, Providence, RI, 1994, p. 15-18.

¹⁰This is the proof reproduced in EGA IV 21.11.1.

Recall that one defines $\det(L)$ as the maximal exterior power of L . In the case under consideration, $\det(L) \simeq L$ since L is invertible, hence the lemma allows one to conclude. Let us prove this lemma. Let

$$0 \longleftarrow L_0 \longleftarrow L_1 \longleftarrow L_2 \longleftarrow \cdots \longleftarrow L_n \longleftarrow 0,$$

be the announced exact sequence, where $L_0 = L$. Since everything is locally free, one has:

$$\bigotimes_{0 \leq i \leq n} (\det(L_i))^{(-1)^i} \simeq \mathcal{O}_X$$

now all the L_i , $i > 0$, are free, hence also their determinants, hence also that of $L_0 = L$. $\square$

We still need to prove (ii) of the theorem. Beforehand, let us prove a lemma which will allow one to proceed by induction:

LEMMA 3.16. *Let A be a noetherian local ring quotient of a regular ring. Let $t \in \mathfrak{r}(A)$ be an A -regular element. Suppose that A is complete for the t -adic topology. Set $X' = \text{Spec}(A)$, $Y' = V(t) \simeq \text{Spec}(B)$, $B = A/tA$, $X = X' - \{x\}$, $Y = Y' - \{x\}$, $x = \mathfrak{r}(A)$. Suppose that:*

- a) *for every $y \in X$ closed in X one has $\text{prof } \mathcal{O}_{X,y} \geq 2$,*
- b) *$\text{prof } A/tA \geq 3$,*

then the map $\text{Pic}(X) \rightarrow \text{Pic}(Y)$ is injective. In particular, if B is parafactorial, A is so as well.

One knows that a) implies $\text{Lef}(X, Y)$ thanks to ?? ??. If one proves that $H^1(Y, P_n) = 0$ for every $n \geq 0$, one will know thanks to (2.2) that $\text{Pic}(X) \rightarrow \text{Pic}(Y)$ is injective. If, moreover, B is parafactorial, one will know that $\text{Pic}(Y) = 0$ (3.5) hence $\text{Pic}(X) = 0$, and $\text{prof}(A) = 3 + 1 \geq 2$ since t is A -regular, hence A will be parafactorial by 3.5.

Let $\mathcal{I} = \widetilde{(tA)}$ be the $\mathcal{O}_{X'}$ -Module associated with the ideal tA . In no. 1, one set $P_n = (\mathcal{I}^{n+1}/\mathcal{I}^{n+2})|_Y$, for every $n \geq 0$. Now t is A -regular hence $P_n \simeq \mathcal{O}_Y$. It therefore remains to prove that $H^1(Y, \mathcal{O}_Y) = 0$. Now $Y = Y' - \{x\}$ is an open of Y' , one therefore has an exact sequence (?? (??)):

$$H^1(Y', \mathcal{O}_{Y'}) \longrightarrow H^1(Y, \mathcal{O}_Y) \longrightarrow H_x^2(Y', \mathcal{O}_{Y'}),$$

whose right-hand term vanishes by virtue of hypothesis b), and whose left-hand term vanishes because Y' is affine. $\square$

LEMMA 3.17. *Keeping the hypotheses of 3.16, suppose moreover that:*

- c) *for every y closed in Y , one has $\text{prof } \mathcal{O}_{X,y} \geq 3$,*
- d) *$\text{prof } A/tA \geq 4$ (stronger than b),*
- e) *for every closed y of X , $y \in Y$, the ring $\mathcal{O}_{X,y}$ is parafactorial.*

Then the map $\text{Pic}(X) \rightarrow \text{Pic}(Y)$ is an isomorphism, in particular in order that A be parafactorial, it is necessary and sufficient that B be so.

One knows (?? ??) that a) and c) imply $\text{Lef}(X, Y)$. Moreover, by the reasoning which has just been given, d) implies that $H^1(Y, P_n) = 0$ for every $n \geq 0$ and $i = 1$ or $i = 2$. Moreover, for every open neighbourhood U of Y in X , the complement of U in X consists of a finite number of closed points. Thanks to e) and to theorem 3.12, one deduces that $\text{Pic}(X) \rightarrow \text{Pic}(Y)$ is an isomorphism. On the other hand $\text{prof } A \geq \text{prof } B \geq 2$; by the criterion 3.5 one deduces that A is parafactorial if and only if B is so.

Let us now prove 3.13 (ii). Let R be a regular noetherian local ring. Let $(t_1, \dots, t_k)$ be an R -sequence. Set $B = R/(t_1, \dots, t_k)$ and suppose that $\dim B \geq 4$. One must prove that B is parafactorial. One proceeds by induction on k . If $k = 0$, B is regular, hence factorial by 3.13 (i), hence parafactorial by 3.10. Suppose $k \geq 1$ and the theorem proved for $k' < k$. Set $A = R/(t_1, \dots, t_{k-1})$ hence $B = A/t_k A$. One may assume B complete by 3.7. By the induction hypothesis, A is parafactorial. Let us prove that one may apply lemma 3.17. One has assumed B complete hence also A , and therefore A is complete for the t -adic topology. If $x \in X$, and if x is closed in X , A_x is a complete intersection of dimension ≥ 4 , with $k' < k$.

By the induction hypothesis A_x is parafactorial, and moreover of depth ≥ 4 . Which gives a), c) and e). Moreover, $\dim A \geq 5$, whence d). $\square$

THEOREM 3.18. *Let X be a locally noetherian prescheme and let $\mathcal{I}$ be a coherent sheaf of ideals of X . Set $Y = V(\mathcal{I})$. Let n be an integer. Suppose that:*

- (i) *one has $\text{Leff}(X, Y)$ (cf. examples ?? ?? and ?? ??),*
- (ii) *for every $p \geq n$, one has $H^i(Y, \mathcal{I}^{p+1}/\mathcal{I}^{p+2}) = 0$ for $i = 1$ and $i = 2$,*
- (iii) *for every open $U \supset Y$ and every $x \in X - U$, the ring $\mathcal{O}_{X,x}$ is regular of dimension ≥ 2 or a complete intersection of dimension ≥ 4 .*

Then, for every open $U \supset Y$ and every integer $p \geq n$, the homomorphisms

$$\text{Pic}(X) \longrightarrow \text{Pic}(U) \longrightarrow \text{Pic}(Y_p)$$

are isomorphisms, where Y_p denotes the prescheme $(Y, \mathcal{O}_X/\mathcal{I}^{p+1})$.

It suffices to combine 3.12 and 3.13.